TITLE: Digital Impression System Usability Specification				
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APPROVALS				
Originator	DPI Engineering	DPI Research & Development	DPI Quality Engineering	DPI Program Management

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Sybron Dental Specialties

1 Introduction

This document describes the Usability Specification for the DPI Digital Impression System, as defined in IEC 62366:2008, 5.5.

2 Inputs

2.1 Application Specification

2.1.1 Intended Medical Indication

The Digital Impression System for Orthodontic Use is an optical impression system intended for use by dental professionals to record the topographical characteristics of patients' teeth, gingiva, and/or palate. The Digital Impression System is intended for use in conjunction with the production of orthodontic appliances, including aligners. It is intended to replace traditional impressions, which are the current standard for recording dental and orthodontic features.

2.1.2 Intended Patient Population

Description – Any orthodontic patient who would benefit from orthodontic treatment, and who needs an accurate impression of his or her teeth in order to produce one or more orthodontic appliances.

Age – 8 years and older. Orthodontic treatment often begins between ages 8 and 14, but adults can also benefit from orthodontic correction

2.1.3 Intended Part of the Body or Type of Tissue Applied to or Interacted With

The tip of the scanning wand will be used in the oral cavity, and will contact the lips, teeth, and mucous membranes within the mouth.

2.1.4 Intended User Profile

Intended end-users for the Digital Impression system are licensed orthodontists and orthodontic clinicians who have been trained by Ormco-qualified trainers in the safe and effective use and maintenance of the system.

2.1.5 Intended Conditions for Use

Intended conditions for use are climate-controlled operatories (10°C to 35°C, 10% to 80% relative humidity) inside orthodontic treatment facilities that use standard dental hygienic procedures. The system is portable, so that it can be moved from operatory to operatory. It is expected that the system will be used daily to record digital impressions.

Each patient scan will require a new disposable wand tip, and the tip support and locking collar must be sterilized between each patient using the guidelines described in the *Digital Impression System User Guide*. These accessories can be purchased from Ormco.

2.1.6 Principles of Operation

The Digital Impression System for Orthodontic Use is an optical impression system intended for use by dental professionals to record the topographical characteristics of teeth, gingiva, and/or palate. The Digital Impression System is intended for use in conjunction with the production of orthodontic appliances, including aligners. It is intended to replace traditional impressions, which are the current standard for recording dental and orthodontic features.

The system is compact and easily transportable from operatory to operatory. It consists of an integrated optical source and camera within a handheld scanning wand. This wand is connected to a data acquisition module and a mini-PC which can be controlled by a monitor directly connected to the device. The wand is small and light-weight for use by chair-side assistants, and the scanning process is fast and simple to use, allowing the scans of both arches and bites to be accomplished rapidly. A digital model of the scanned areas is displayed in real time on the computer monitor. The operation of the wand is controlled by a software interface which the clinician uses to control the scanning process.

2.2 Primary Operating Functions

2.2.1 Frequently Used Functions

- Clean and disinfect system components (between patients)
- Change/replace the tip support and disposable tip.
- Steam-sterilize the tip support and locking collar.
- Power-on/power-off the system
- Move the system
- Change the touchscreen position
- Touch to select commands on the touchscreen
- Place the wand in the wand holder
- Remove the wand from the wand holder
- Place barriers on system components
- Remove barriers from system components
- Position the wand in the patient's mouth
- Move the wand in the patient's mouth while scanning
- Scan lower arch
- Scan upper arch
- Scan bite(s)
- Run a Fill scan segment
- Run a system check (at power-on)
- Add/remove/edit users
- Add/edit practice information
- Create a new case and enter patient information
- Use the touchscreen to rotate images
- Use the image tools to inspect scan segments
- Cancel/redo a scan segment
- Use Skip to skip a scan segment
- Complete and approve a digital impression

2.2.2 Functions Related to Safety

- Clean and disinfect system components (between patients)
- Change/replace the tip support and disposable tip.
- Steam-sterilize the tip support and locking collar.
- Power-on/power-off the system
- Move the system
- Change the touchscreen position
- Place the wand in the wand holder
- Remove the wand from the wand holder
- Place barriers on system components
- Remove barriers from system components
- Position the wand in the patient's mouth
- Move the wand in the patient's mouth while scanning
- Scan lower arch
- Scan upper arch
- Scan bite(s)

2.3 Things that Could Go Wrong

Sources: expert (dental clinician), risk analysis (QA/QE staff).

2.3.1 During Normal Use

- Power failure

2.3.2 Use Errors

- Failure to properly clean and disinfect system components between patients
- Failure to properly steam sterilize the tip support and locking collar between patients
- Failure to replace the disposable tip between patients
- Tip support and locking collar not installed properly
- Disposable tip not installed properly
- Applying excessive force to system components
- Dropping system components
- Tipping over base unit by applying excessive force to extended touchscreen
- Mechanical stress (sitting or stepping on wand or other system components)
- Improper use of barriers
- Incorrect placement and movement of wand in patient’s mouth
- Insufficient scan data collected
- Operator stares into the laser beam emitted by the wand, or views the beam directly with an optical instrument.

2.3.3 Environment

- Overheating
- Liquids penetrating system components

2.3.4 Patient

- Bites on wand tip
- Puncturing/gouging tissue in patient's mouth (with wand tip)

2.3.5 Hygiene

- Cross-contamination between patients

2.3.6 Application

- Wand not positioned and used properly
- Wand used outside of the oral cavity
- Measurements (scans) not performed correctly

2.4 Hazards and Hazardous Situations Related to Usability

2.4.1 Characteristics Related to Safety

This section lists system characteristics of the Digital Impression System related to safety that pertain to usability, and as described in ISO 14971:2007, 4.2, as required by IEC 62366:2008, 5.3.1.

The following characteristics listed in the Digital Impression System Risk Management files are related to safety and also pertain to usability:

- #3 – The system is in contact with patients and users.
- #5 – Energy is delivered to the patient.
- #9 – System components are intended to be routinely cleaned and disinfected.
- #11 – Measurements are taken of the patient's teeth.
- #17 – There are consumables associated with the system.
- #18 – Maintenance and calibration are necessary.
- #19 – Software is used to operate the system.
- #26 – Training is required to operate the system.
- #29.1 – UI design features can contribute to use error.
- #29.3 – The device has connection parts/accessories.
- #29.4 – The device has a control interface.
- #29.5 – The device displays information.
- #29.6 – The device is controlled by a menu.
- #29.8 – The UI can be used to initiate user actions.
- #33 – The device is portable.
- #34 – The use of the device depends on essential performance.

2.4.2 Known or Foreseeable Hazards

This section identifies the known or foreseeable hazards and hazardous situations related to safety that pertain to usability for the DPI Digital Impression System, as defined in IEC 62366:2008, 5.3.2.

2.4.2.1 Hardware

The following are the known or foreseeable hazards listed in the Digital Impression Risk Management File (Hardware) that are related to safety and also pertain to usability:

- #H006 – Laser eye safety.
- #H012 – Pinch points on touchscreen arm
- #H013 – Too much force on touchscreen could cause unit to tip over
- #H016 – Dropped wand could cause sharp edges
- #H022 – Bio-contamination: wand body
- #H023 – Bio-contamination: wand body
- #H024 – Bio-contamination: wand tip
- #H025 – Bio-contamination: wand tip
- #H026 – Bio-contamination: base unit
- #H027 – Bio-incompatibility: wand body
- #H028 – Electrical hazard if cleaning solutions penetrate base unit enclosure.
- #H033 – Skin irritation
- #H037 – Inappropriate instructions for use
- #H038 – Inadequate description of performance characteristics
- #H039 – Inadequate description of intended use
- #H040 – Inadequate disclosure of limitations
- #H041 – Inadequate specification of accessories
- #H042 – Inadequate specification of pre-use checks
- #H043 – Overly-complicated operating instructions
- #H044 – Inadequate specification of service and maintenance
- #H045 – Inadequate hazard warnings
- #H046 – Inadequate instructions for use
- #H047 – Inadequate operator training
- #H048 – Disposable tip falls off
- #H049 – Disposable tip mounted incorrectly
- #H050 – Disposable tip reused
- #H052 – Disposable tip cracked or broken
- #H053 – Dropped wand could deform wand tip
- #H054 – Dropped wand could cause sharp edges
- #H055 – Wand tip support not sterilized
- #H056 – Wand tip support not installed correctly
- #H057 – Wand tip support not used
- #H059 – Wand tip support cracked or broken
- #H060 – Misuse of the wand/laser light exposure
- #H062 – Wand too unwieldy
- #H065 – Cord wrapped improperly
- #H071 – Failure to disinfect wand body

- #H075 – Failure to disinfect base unit

2.4.2.2 Software

There are no known or foreseeable hazards listed in the Digital Impression Risk Management File (Software) that are related to safety and also pertain to usability.

2.5 Preliminary Review of the User Interface Concept

2.5.1 Design Requirements

As described in SP1-4000, Product Performance Specification, "Voice of the Customer" (VOC) requirements data for both hardware and software components was collected from orthodontists and chair-side clinicians over a period of four months, and the results posted in the Design History File.

Based on the Voice of the Customer data, five general areas of user need were determined for the Digital Impression System:

- Speed – fast scanning to compete with conventional impression taking
- Accuracy – to capture whole-arch data with low error
- Ease-of-use – to minimize ergonomic stresses and specialized training
- Durability – to withstand operatory handling and transport
- Safety – to ensure patient safety and general comfort

These general requirements were further broken down into specific user needs – including usability and UI requirements – and the specific UI requirements were listed in detail in the Product Performance Specification.

2.5.2 Design Process

2.5.2.1 Introduction

The UI design was an iterative process, with several design and development activities taking place concurrently:

- Detailed listing of product requirements in the Product Performance Specification, including usability and patient safety requirements.
- Detailed description of the UI requirements in the Hardware Requirements Specification and the Software Requirements Specification, including workflow charts.
- Risk Analysis per ISO 14971:2007, Risk Management.
- UI design and review.
- Production and testing of UI features on preliminary versions of the system hardware and software.
- Implementation of a feature-complete Graphical User Interface (GUI) in a Beta version of the software.

2.5.2.2 GUI Requirements in the Software Requirements Specification

One of the first design outputs based on the Product Performance Specification (PPS) was the Software Requirements Specification (SRS).

The SRS was developed over a period of several months. Several preliminary versions of this document were produced, with each subsequent version further detailing the requirements specified in the PPS. An initial version of the SRS was produced on 4-10-11, and a complete draft version – including the GUI specifications – was completed on 8-10-11. These documents are available for review in the Device History Record.

The SRS includes a detailed diagram of the user workflow, and describes in detail the GUI screens required for each step in the workflow. The GUI section of the SRS also includes a table detailing each interface feature screen-by-screen, with each feature referenced by a unique Software Requirements number.

2.5.2.3 GUI Screen Design

The GUI screens were designed in parallel with the SRS. This was also an iterative process involving multiple screen design and review cycles.

People Centered Design (PCR)

PCR, a human factors research and design vendor, reviewed the available requirements, then generated preliminary workflow diagrams outlining the primary operating functions of the GUI. PCR also generated mock-ups of preliminary GUI screen designs.

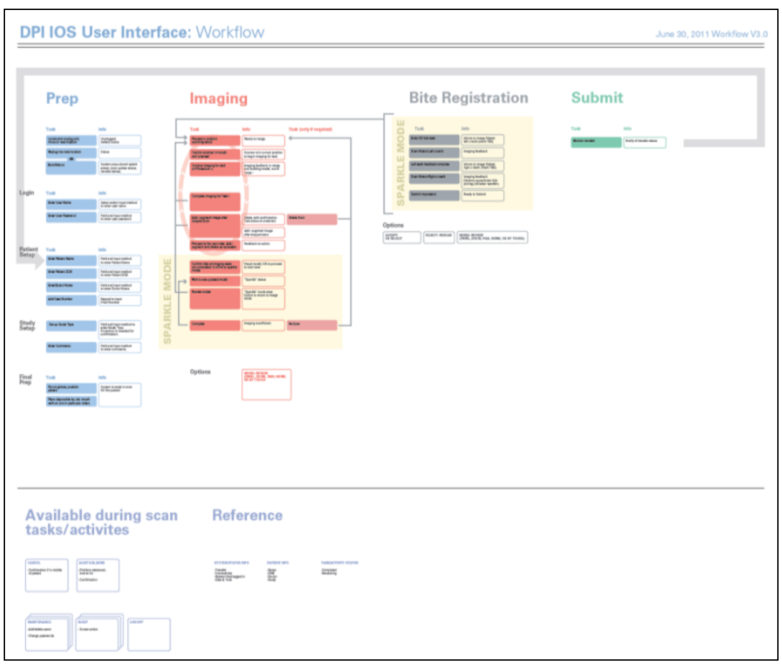

Figure 1: Preliminary GUI Operating Functions -- PCR

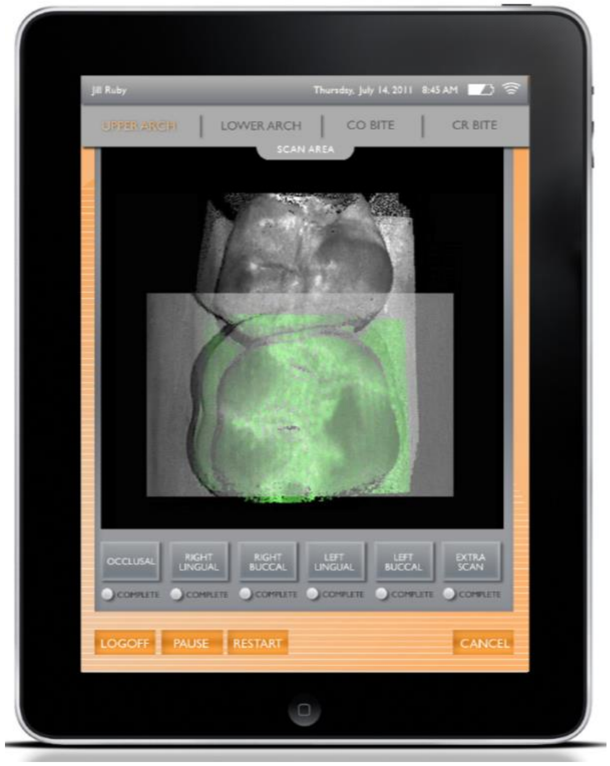

Figure 2: Preliminary Screen Design -- PCR

DPI engineering personnel reviewed the PCR designs and incorporated portions of this design output into the overall design strategy. Records of the interactions with PCR are available in the DHR.

Usability Pro

Usability Pro, another design vendor, reviewed the DPI requirements and preliminary design outputs, as well as the PCR design outputs. Usability Pro further refined the GUI design, and produced detailed Photoshop drawings of screen designs. Usability Pro also produced a preliminary style guide detailing color, fonts, button design, and sizes of all elements to be used in the GUI. The design and placement of screen elements -- such as command buttons, menus, labels, and icons -- was based on commonly accepted principles of usability and user interface design.

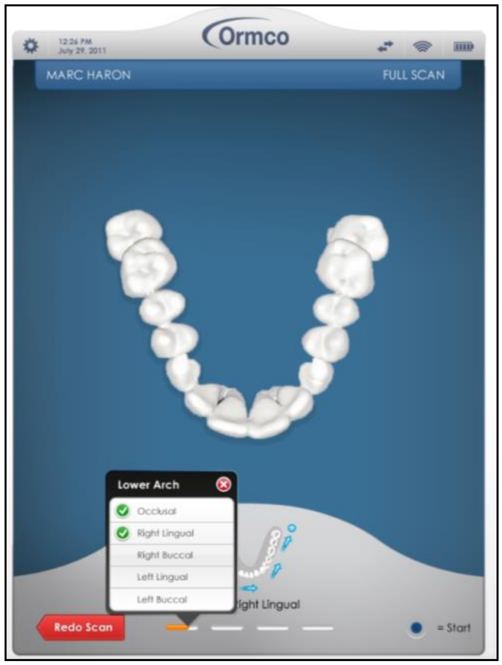

Figure 3: Usability Pro GUI Screen

DPI engineering personnel reviewed the Usability Pro designs and incorporated portions of this design output into the overall design strategy. The Usability Pro design outputs are available in the DHR.

Preliminary UI Software and Hardware

GUI workflow and design elements were incorporated into a preliminary version of the device software known as “Topeka,” which was installed in a preliminary version of the system hardware. The system hardware included prototype scanning wands.

Use Scenarios and UI elements were tested by DPI designers, product management, and a dental clinician using the Topeka version of the software and a preliminary version of the system hardware.

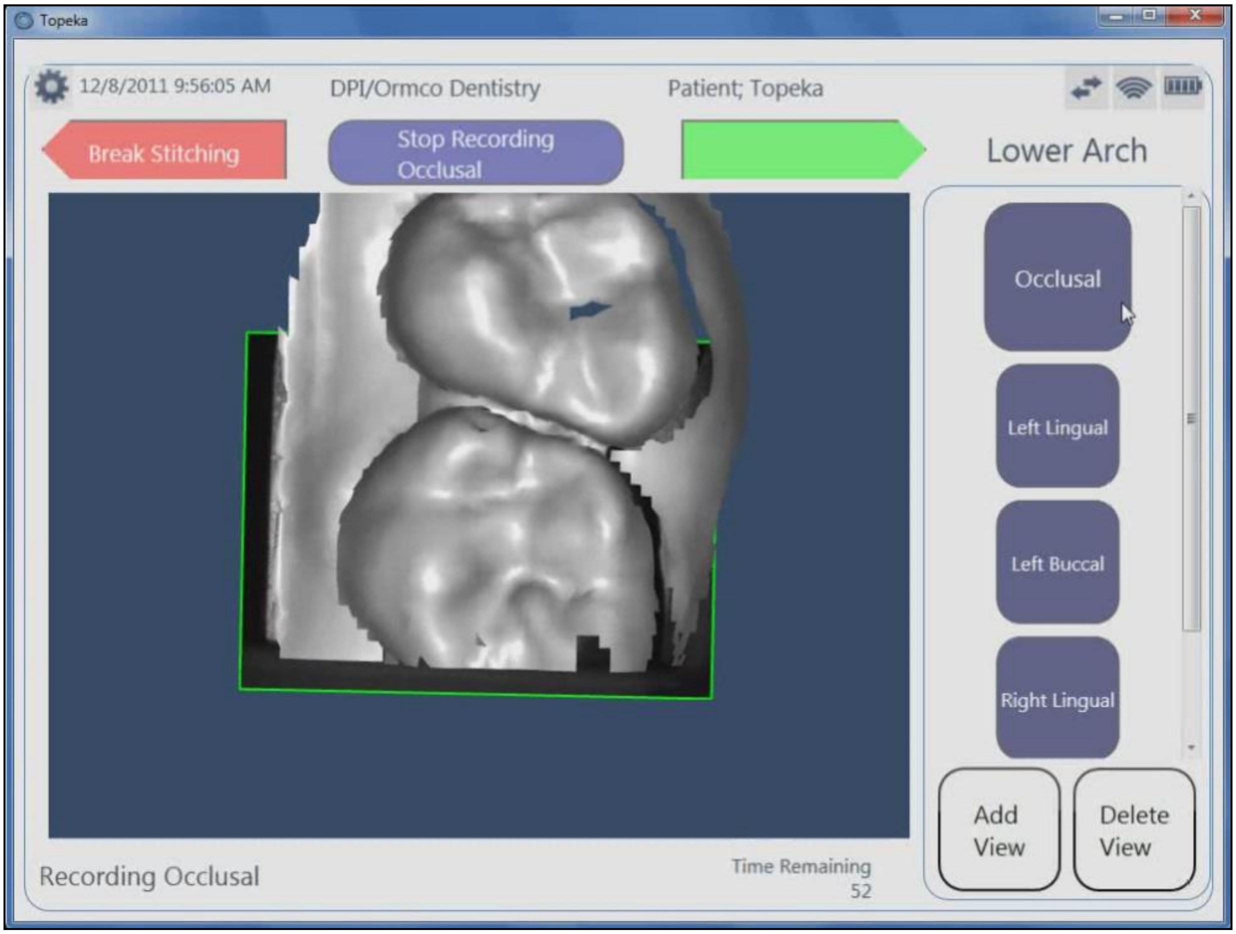

Figure 4: Topeka Version of GUI

Design outputs were adjusted based on this testing – and the PPS and SRS requirements – and iteratively refined with subsequent versions of the Topeka software.

2.5.3 Preliminary Review of the User Interface Concept

a. Clean and disinfect system components.
Conclusion: Operators will need to be trained to properly clean and disinfect system components

b. Replace the tip support and the disposable tip.
Conclusion: Operators will need to be trained to replace the tip support and the disposable tip.

c. Steam sterilize the tip support and locking collar.
Conclusion: Steam sterilization parameters need to be documented.

d. Power-on/power-off the system.
Conclusion: no issues.

e. Run a System Check.
Conclusion: System Check will be required each time the system is powered-on. Procedures for what to do if System Check fails need to be documented.

f. Move the system.
Conclusion: no issues.

g. Adjust the touchscreen position.
Conclusion: Swivel capability may need to be added to touchscreen.

h. Select commands on the touchscreen.
Conclusion: Touchscreen sensitivity may need to be adjusted.

i. Place the wand in the wand holder.
Conclusion: no issues.

j. Remove the wand from the wand holder.
Conclusion: no issues.

k. Place barriers on system components.
Conclusion: Operators will need to be instructed (in documentation and training) not to place a barrier over the opening in the tip of the scanning wand, or over ventilation slots.

l. Remove barriers from system components.
Conclusion: no issues.

m. Position the wand in the patient's mouth.
Conclusion: Operators will need to be trained to properly position the wand in the patient’s mouth.

n. Move the wand in the patient's mouth while scanning.
Conclusion: Operators will need to be trained to properly move the wand in the patient's mouth.

o. Create a new case and enter patient information.
Conclusion: Operators will need to be trained to create a new case and enter patient information.

p. Scan the patient's lower arch.
Conclusion: Operators will need to be trained to successfully scan the patient's lower arch.

q. Scan the patient's upper arch.
Conclusion: Operators will need to be trained to successfully scan the patient's upper arch.

r. Run left and right bite scans.
Conclusion: Operators will need to be trained to successfully run bite scans.

s. Run a Fill scan.
Conclusion: Operators will need to be trained to successfully run fill scans
.

t. Skip a scan segment.
Conclusion: Operators will need to trained on the implications of skipping a scan segment.

u. Cancel and repeat a scan segment.
Conclusion: Operators will need to be trained to cancel and repeat a scan segment.

v. Use the touchscreen and image tools to inspect images.
Conclusion: Operators will need to be trained to inspect images, including acceptance criteria for individual scans (lower arch, upper arch, and bite scans) and completed digital impressions.

w. Complete and approve a digital impression.
Conclusion: Operators will need to be trained regarding acceptance criteria for completed digital impressions.

x. Add, edit, or remove users.
Conclusion: Operators will need to be trained to add, edit, and remove users.

y. Add or edit practice information.
Conclusion: Operators will need to be trained to add or edit practice information.

3 Usability Specification

3.1 Use Scenarios

3.1.1 Preliminary Use Scenarios

- Trained operator properly prepares system for use in dental operatory
- Operator enters patient information
- Operator scans patient's teeth
- Operator completes and approves the digital impression

3.1.2 Worst-Case Use Scenarios

Use of the Digital Impression System is restricted to dental clinicians who have been formally trained and certified by Ormco-approved trainers. Use of the system is also restricted to dental operatories. Therefore, the Preliminary Use Scenarios listed in the preceding section represent the worst-case Use Scenarios for the system.

3.2 User Actions Related to Primary Operating Functions

- Operator cleans and disinfects system components between each patient.
- Operator replaces the tip support and the disposable tip between patients.
- Operator steam-sterilizes the tip support and locking collar between patients.
- Operator powers-on/powers-off the system.
- Operator runs a system check (at power-on).
- Operator moves the system.
- Operator adjusts the touchscreen position.
- Operator selects commands on the touchscreen.
- Operator places the wand in the wand holder.
- Operator removes the wand from the wand holder.
- Operator place barriers on system components.
- Operator removes barriers from system components.
- Operator positions the wand in the patient's mouth.
- Operator moves the wand in the patient's mouth while scanning.
- Operator creates a new case and enters patient information.
- Operator scans the patient's lower arch.
- Operator scans the patient's upper arch.
- Operator runs left and right bite scans.
- Operator runs a Fill scan.
- Operator skips a scan segment.
- Operator uses the touchscreen to inspect images.
- Operator completes and approves a digital impression.
- Operator adds/edits/removes users.
- Operator adds/edits practice information
- Operator cancels and repeats a scan segment.

3.3 User Interface Requirements for the Primary Operating Functions

Note: Further details regarding hardware and software User Interface requirements are listed in the *Digital Imaging System Product Performance Specification*.

- UI01 – Easy to clean and disinfect system components between each patient as defined in 149-0901 *Digital Impression System User Guide*.
- UI02 – Easy to replace and securely attach the tip support and the disposable tip as defined in 149-0901 *Digital Impression System User Guide*.
- UI03 – One-button power-on/power-off.
- UI04 – System check is performed when the system is powered-on (before the first scan).
- UI05 – System is lightweight enough to move easily.
- UI06 – Touchscreen position is easily adjustable.
- UI07 – Easy to select commands on the touchscreen.
- UI08 – Easy to remove the wand from the wand holder.
- UI09 – Easy to safely seat the wand in the wand holder.
- UI10 – Wand tip fits comfortably in the patient's mouth.
- UI11 – Wand is lightweight to enable the operator to move it easily while scanning.
- UI12 – Easy to create a new case and enter patient information.
- UI13 – Operator can successfully scan the patient's lower arch.
- UI14 – Operator can successfully scan the patient's upper arch.
- UI15 – Operator can successfully run left and right bite scans.
- UI16 – Operator can successfully run a Fill scan.
- UI17 – Operator can successfully cancel and repeat a scan segment.
- UI18 – Operator can successfully skip a scan segment.
- UI19 – Operator can use the touchscreen to inspect images.
- UI20 – Operator can successfully complete and approve a digital impression.
- UI21 – Operator can add, edit, and remove users (must be logged in as an Admin).
- UI22 – Operator can add or edit practice information (must be logged in as an Admin).
- UI23 – Operator understands that he/she should not stare at the laser beam emitted by the wand, or directly view the beam with an optical device.

3.4 Acceptance Criteria for User Interface Requirements for the Primary Operating Functions

For all of the requirements listed in the previous section (3.3), acceptance criteria will consist of observation of operations by an Ormco trainer, Quality Assurance Engineer, or Product Manager who is an established expert in the operation of the Digital Impression System.

There are additional acceptance criteria for UI20. Traditional impressions will be taken of each patient, and used to manufacture stone models. Five orthodontists will review each digital impression and render their professional opinion on the quality of the digital impression. For comparison purposes, the stone model made from the traditional impression for each patient will be provided. A digital impression will be considered acceptable after review and acceptance by a majority (three or more) of the five orthodontists.

3.5 Issues to be Addressed by Training, Documentation, and Labeling

- Overheating – operating environment, blocking ventilation openings
- Cross-contamination between patients (use proper cleaning/disinfecting/sterilization procedures)
- Improper cleaning/disinfecting procedure leads to liquids penetrating system components.
- Improper replacement of tip support and locking collar (tip support falls off).
- Improper replacement of disposable tip (disposable tip falls off).
- Wand not placed safely in wand holder when not in use.
- Wand not positioned properly in patient's mouth.
- Operator cannot create a new case and enter patient information.
- Operator cannot successfully scan the patient's lower arch.
- Operator cannot successfully scan the patient's upper arch.
- Operator cannot successfully run left and right bite scans.
- Operator cannot successfully run a Fill scan.
- Operator cannnot successfully cancel and repeat a scan segment.
- Operator cannot successfully skip a scan segment.
- Operator does not know how to use the touchscreen to inspect images.
- Operator cannot successfully complete and approve a digital impression that can be verified by Insignia as sufficient to complete a doctor's order for a standard twin bracket case.
- Operator does not know how to add, edit, and remove users.
- Operator does not know how to add or edit practice information
- Operator stares into the laser beam emitted by the wand, or views the beam directly with an optical instrument.

Note: Training, documentation materials will be designed in accordance with the intended use and user profiles listed in the Application Specification in Section 2.1 of this document.

3.6 User Interface Requirements for those Use Scenarios that are Frequent or Related to Safety

The following procedures will be easy for Operators to perform after undergoing training and certification by Ormco-certified trainers.

- Easy to clean and disinfect system components between each patient.
- Easy to replace and securely attach the tip support and the disposable tip.
- One-button power-on/power-off.
- System check is performed when the system is powered-on.
- System is lightweight enough to move easily.
- Touchscreen position is easily adjustable.
- Easy to select commands on the touchscreen.
- Easy to remove the wand from the wand holder.
- Easy to safely seat the wand in the wand holder.
- Wand tip fits comfortably in the patient's mouth.
- Wand is lightweight to enable the operator to move it easily while scanning.
- Easy to create a new case and enter patient information.
- Operator can successfully scan the patient's lower arch.
- Operator can successfully scan the patient's upper arch.

- Operator can successfully run left and right bite scans.
- Operator can successfully run a Fill scan.
- Operator can successfully cancel and repeat a scan segment.
- Operator can successfully skip a scan segment.
- Operator can use the touchscreen to inspect images.
- Operator can successfully complete and approve a digital impression.
- Operator can add, edit, and remove users.
- Operator can add or edit practice information

3.7 Requirements for Determining Whether Primary Operating Functions are Easily Recognizable by the User

No further requirements beyond those already listed in User Interface Requirements for the Primary Operating Functions.

4 Revision History

Revision	Date	Description Of Change	Author
1.0	6-11-12	Draft version	D. Hoyle